The Spacial Analysis of Crime Clustering

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1 Source Paper

The original paper, titled *Social Justice Implications of Model Selection for Point Process Models of Crime* (Clark & Mateu, 2026), analyzed the clustering of homicides in Chicago in 2015 and the reasons behind why that may be. The current widespread assumption in criminology is that crime spreads through a contagion-like process. That is, the clustering of crime arises from offenders learning which targets are easily victimized through the actions of other criminals, therefore increasing the probability that another crime in that area will occur. The paper researched the modeling of spatio-temporal clustering using the Log-Gaussian Cox Process as well as the Hawkes Process.

The Log-Gaussian Cox Process is a way to spatially model the clustering. This process is normally used for point processes and allows us to analyze interactions through environmental factors. In order to define the Log-Gaussian Cox Process, it was necessary to define the functions of the observed factors and the functions of the unobserved factors. The results of the process are shown below, in which we can see a clear spatio-temporal clustering.

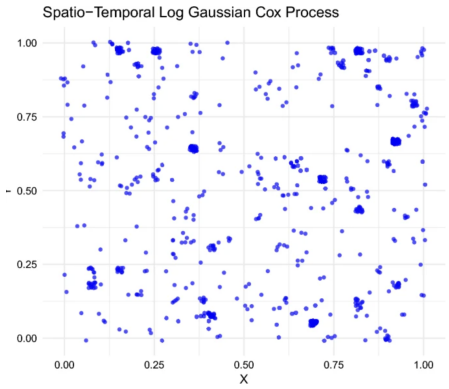


Figure 1: A Log-Gaussian Cox Process with constant expectation

On the other hand, the Hawkes Process is used to analyze "copy cat" crime. That is, investigating whether or not one arrival into the system increases the probability of another arrival into the system. It is a self-exciting point process that models clustered, dependent events based on conditional intensity function $\lambda(x)$ that jumps at each event. The results of the Hawkes Process described in the article are shown below.

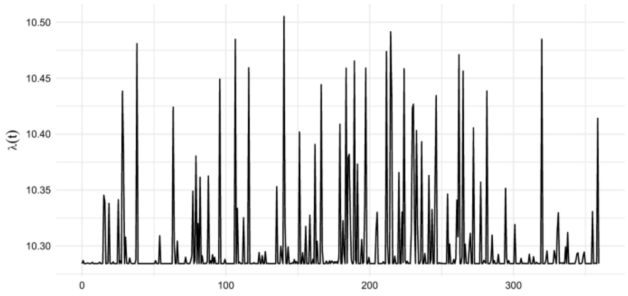


Figure 2: Plot of the intensity of the spatio-temporal Hawkes process over time

Based on the data and analysis, it was unclear whether the repeat victimization was due to underlying small-scale spatial or temporal external factors. The Hawkes Process was therefore not an efficient method of analyzing clustering in this experiment because the initial assumptions needed to be analyzed in order to use it. In particular, the crime detection methods, origin, and availability of the data were all unknown to the researchers.

The code for the project is included in a single R file, titled `Chapter4Code.R` in the Github. The R file contains the code for the Log-Gaussian Cox Process and the Hawkes Process with their associated plots.

Overall, the paper concluded that the Log-Gaussian Cox Process was an efficient clustering metric, and that clustering of crime did exist in this dataset. The paper related the results of the process to current criminology theories and concluded that the mathematical clustering of crime must be due to socioeconomic and social justice issues, disproving the assumption that the clustering is a contagion-like process.

2 Project Research

Our research was concerned with spatially analyzing the clustering of crime in different cities. In particular, we wanted to know how more or less "crime-prone" one area would be over another. Using these results, we also wanted to know what kind of crime that area would be more or less prone to. Finally, we wanted to understand the types of functions and point processes that were best for crime and clustering analysis to form our conclusions. The cities that we analyzed were Chicago, Vancouver, and New York City.

2.1 Numerical Methods and Results

The numerical methods employed were the K-function, Stationary and Linear Poisson Processes, the Page Rank Algorithm, the Graph Convolutional Network, and Concave Hulls.

2.1.1 Stationary v. Linear Poisson Point Process

The purpose of the K-Function was to test for spatial randomness. In particular, a high $K(r)$ value for a small distance r implies the existence of clustering. Our research explored if a cubic function fitted to an inhomogeneous Poisson point process matches crime clustering, or lack thereof. Our plots modeled a homogeneous process (no clustering) and an inhomogeneous point process (clustering). We see clustering when there is a difference between the theoretical and actual $K(r)$ values.

The polynomial used to model the data was of degree 3. A degree 1 polynomial would imply that crime is monotonic in a certain direction. A degree 2 polynomial would imply that crime only has one "hotspot" in the spatial data. The cubic function was of the form

$$x + y + x^2 + xy + y^2 + xy^2 + x^2y + x^3 + I(y^3)$$

and was therefore used because a polynomial of degree 3 implies the existence of multiple "hotspots" in the spatial data. The results of the modeling for each city are shown below.

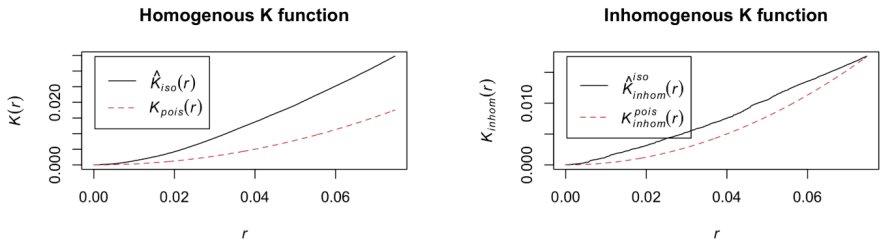


Figure 3: K-Function results for Chicago with mean difference 0.0079 (left) and 0.0018 (right)

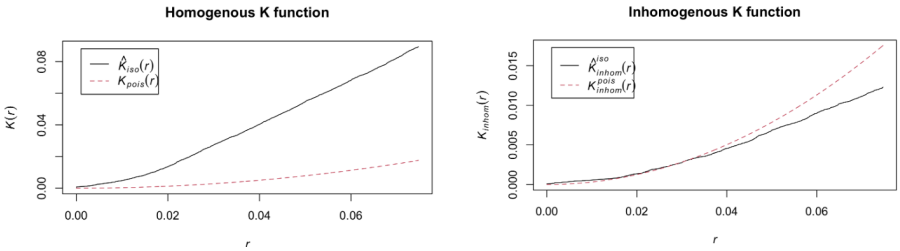


Figure 4: K-Function results for Vancouver, BC with mean difference 0.033 (left) and -0.001 (right)

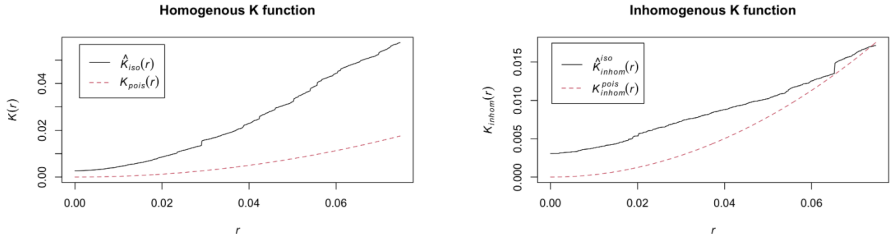


Figure 5: K-Function results for New York City with mean difference 0.018 (left) and 0.0008 (right)

The results of the point process implied that clustering of crime exists in each of these cities. In Figure 3, it is shown that Chicago better fits the inhomogeneous model, implying the existence of crime clustering. Similarly with Vancouver in Figure 4, we can see that since the theoretical and actual $K(r)$ values differ so heavily in the homogeneous model, then the distribution of crime is most likely inhomogeneous implying the existence of "safe" and "unsafe" areas. We can also see that our cubic function fits the inhomogeneous model very well for Vancouver. Finally, in Figure 5 we see that the homogeneous model for New York City has closer theoretical and $K(r)$ values (lower mean difference), implying that although clustering exists, the spread of crime is slightly more uniform that that of Vancouver.

2.1.2 Page Rank Algorithm

Analyzing crime clustering based on the Page Rank Algorithm relied on the k-Nearest-Neighbors search. The function worked by taking the k nearest, in our case $k = 10$, neighbors to each point. This was done by calculating the Euclidean distance between each point and each other point. The k nearest neighbors with the smallest distances are then aggregated and chosen. This allowed us to characterize locations where the relative importance of a location

was balanced by the importance of its spatial neighbors. Using the k-Nearest Neighbors search, we were then able to build an adjacency matrix A which was normalized to the transition matrix T. Using the algorithm, this meant that every entry $T[i, j]$ represented the probability of a random walk from point i to point j. Finally, we performed the Page Rank Algorithm using T. The results of the Page Rank Algorithm are shown below.

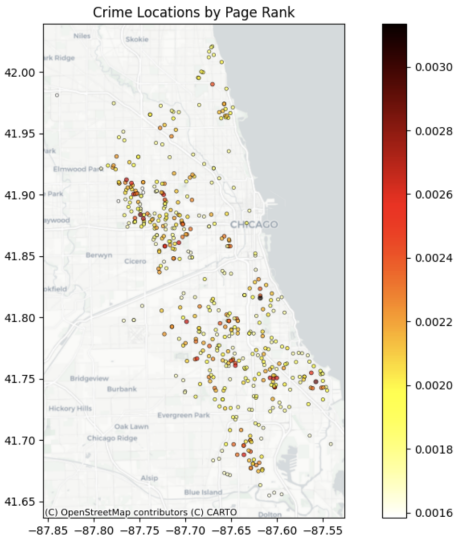


Figure 6: Page Rank Algorithm for Chicago Murder Locations

In the graph, the darker areas represent "hotter" spots for crime whereas the lighter areas represent "colder" spots for crime. Using the Page Rank Algorithm in this context represents the flow of people on city streets. That is, clustering could be from some areas being more "spatially reachable" than others based on its accessibility. Using the algorithm, we can see that our results are consistent with the results from the paper using the Log-Gaussian Cox Process.

2.1.3 Graph Convolutional Network

We then analyzed the types of crime occurring in different cities using a Graph Convolutional Network. This process assumes that crime at a node is a result of what is at that node *and* what is nearby. The network processes data by taking a weighted average of the nodes. That is, if certain types of crime were more clustered in one area, then that type would be weighted higher than others of the same type elsewhere. The network then uses feature aggregation and the learned weights to find the relevant feature combinations.

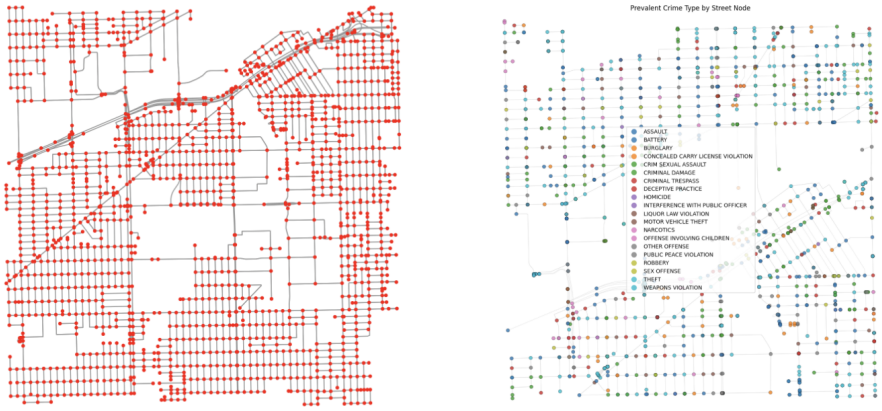


Figure 7: Graph Convolutional Network for Chicago Crime Types

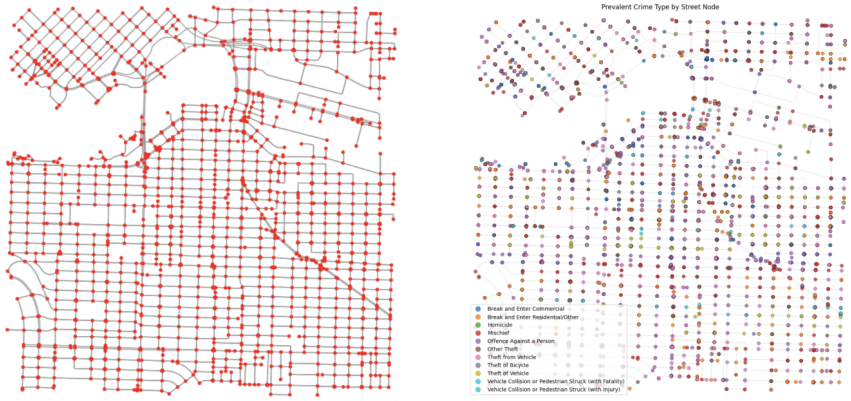


Figure 8: Graph Convolutional Network for Vancouver Crime Types

In Figures 7 and 8, the graphs on the left represent the street node graphs where every node is a point of interest. That is, every node represents an intersection or endpoint where a crime is likely to occur. In the figures, the graphs on the right represent the same street nodes after the Graph Convolutional Network is implemented to find the most common crime type at each node.

As we can see from the Figures, since Vancouver is considered safer than Chicago, the most common crime types at each node are Vehicle Theft and Mischief over violent crimes such as Assault and Homicide.

2.1.4 Concave Hull

Finally, we analyzed the clustering of crime in different cities using Concave Hulls. Concave Hulls were used instead of Convex Hulls in order to not rule out

the entire city.

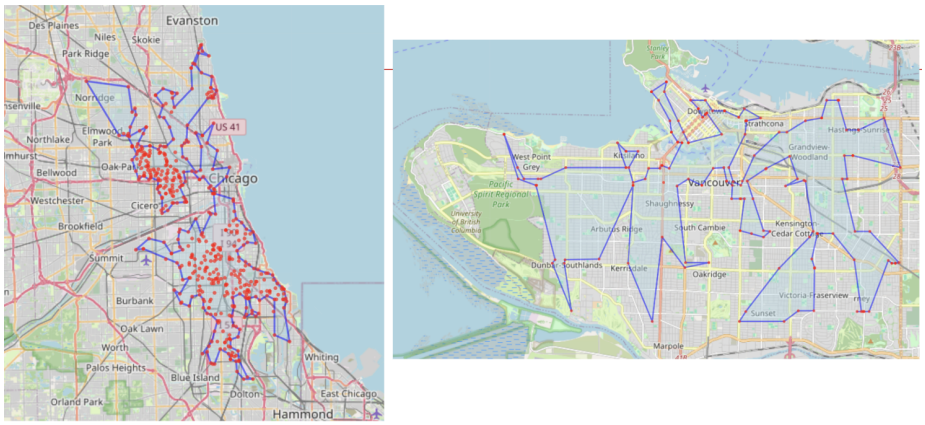


Figure 9: Concave Hulls for Chicago (left) and Vancouver (right)

As we can see from Figure 9, there are areas in both regions that could be considered safe. Using the Concave Hulls we can also see that the areas that would be considered unsafe in Chicago match the clustering patterns outlined in the original paper using the Log-Gaussian Cox Process.

2.2 Github

Our associated Github repository includes our datasets of interest along with the Python and R code used to analyze the datasets. The Chicago homicide dataset used in the original paper is `ChicagoCrimeData.csv`. The columns of interest in this dataset were `longitude` and `latitude`.

Moreover, `crimes_2018.csv` is a dataset that shows all types of Chicago crimes, `vandcrime.csv` is a dataset that shows all types of Vancouver crimes, and `vanhomi.csv` is a dataset that shows Vancouver homicides.

The Jupyter notebook that analyzed the datasets using the Page Rank Algorithm and the Graph Convolutional Network is titled `CrimeDataNotebook.ipynb`.

The analyses using the K-Function and the Concave Hull are done in `vancouver.Rmd` and `nyccode.Rmd`.

2.3 Implications

The use case for our research is mostly predictive policing. Knowing which areas are more "prone" to certain types of crime could help to allocate government and city funds to the correct areas. Our research could also be used to implement evidence-based patrolling independent of bias, leading to overall improved public safety in high crime areas.

3 Challenges and Future Work

One of the main challenges that we faced with our research was forming the relevant questions. We wanted to know how our data could be helpful as well as what the practical applications were. It was also difficult to figure out ways in which we could improve upon the original research. That is, clustering had already been established and we wanted to know if it was worth it to confirm or deny the original research.

One avenue for future research was investigating where exactly these crimes were occurring. In particular, if the clustering was occurring at banks, supermarkets, schools, etc. If we had had access to this data, it could lead to more precaution in these areas and more accurate allocation of city funding. It also would have been interesting to look into the current implementation of government programs and its effects. Finally, time permitting we would have liked to look into the relationship between homicide and other types of crime. More specifically, if increased homicide rates in one area would also imply increased petty crimes in the same area.

4 References

Clark, N., Mateu, J. Social Justice Implications of Model Selection for Point Process Models of Crime. La Matematica 5, 10 (2026). https://doi.org/10.1007/s44007-025-00177-7

https://github.com/nick3703/SocialJusticeCrime/tree/main

https://github.com/kenedyducheine/statistical-crime-analysis

https://www.kaggle.com/datasets/chicago/chicago-crime/data